

Name: _____

Student ID: _____

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?